Study Breaks and Perceived Learning

Exploratory Data Analysis

STA304H5F: Surveys, Sampling and Observational Data
Fall 2026 October 24, 2026
Overleaf Source: `https://www.overleaf.com/project/6add5ee7cbee7aeb664ee9a2`

1 Study design and data preparation

This report follows the latest group proposal. Its three planned analyses are:

1. **RQ1: Chi-square test.** Is break count per session associated with agreement that breaks improve efficiency in the next study cycle?
2. **RQ2: Simple linear regression.** Is break duration linearly associated with perceived retention?
3. **RQ3: One-way ANOVA.** Does mean perceived retention differ across break-activity profiles?

The new CSV contains **57 records and 13 columns**, dated October 5–9, 2026. The actual instrument measures **next-day retention on a 1–10 self-rating**, rather than the proposal's expected immediate test score. Break duration is already numeric in minutes, so no category midpoint is invented. Efficiency is agreement with a benefit statement (1–7), not measured productivity. The report retains the proposal's methods while stating these measurement differences.

Sampling. The proposal plans SRS from collected course responses, with the survey distributed through a Google Form exchange. No completed random-draw procedure is documented. This analysis uses the screened supplied responses; voluntary participation and selection limit generalization. Randomly sampling the respondent pool would not remove its initial self-selection.

Table 1. Preparation rules and analysis denominators.

Issue	Action
Unavailable course IDs	Exclude records 8 and 38, whose IDs are missing in the new CSV. Other IDs have not been verified against the course roster.
Inconsistent session times	Exclude records 12, 17, 30, 33, 43, 48 and 50: break count times duration meets or exceeds total session duration.
Weekly-hour outliers	Set four impossible values (900, 2,800, 600, 600) to missing in this field only. A supplied unresolved range is already missing.
Undefined break duration	Set duration to missing for both zero-break respondents, including one reporting 60 minutes despite zero breaks. One additional duration is already missing from an unresolved range; no midpoint is imputed.
Analysis sizes	57 source records; 9 record exclusions; 48 screened records; 46 break takers; 45 complete regression cases; 43 usable weekly-hour values.

The source CSV is unchanged. Timing flags are conservative analysis exclusions, not proof that the responses are invalid; this screening choice can affect results. Record 25 reports a 780-minute session and 120-minute breaks. It is unusual but internally possible and is retained in the primary fit. Grade values are band proxies, not exact marks. IDs, timestamps and quality notes are administrative fields. Source-record numbers in this report count from 1 after the CSV header. Appendix B visualizes the timing and weekly-hour checks.

2 Summary statistics

Table 2. Numeric summaries of the screened sample.

Measure	n	Missing	Mean	Median	SD	IQR	Min	Max
Study session duration (min)	48	0	150.62	120.00	109.44	60.00	30.00	780.00
Breaks per study session	48	0	2.35	2.00	1.64	2.00	0.00	8.00
Duration of each break (min)	45	3	16.89	10.00	18.81	7.50	5.00	120.00
Next-day retention rating (1-10)	48	0	6.15	6.00	1.40	1.00	3.00	9.00
Weekly study time (hours)	43	5	15.81	15.00	9.61	12.00	5.00	40.00
Prior grade: band proxy (%)	44	4	74.77	75.00	8.49	7.50	55.00	95.00
Break benefit agreement (1-7)	48	0	5.73	6.00	1.16	1.00	2.00	7.00

Generated with `datawizard::describe_distribution()` from the easystats ecosystem. SD is the sample SD; IQR uses datawizard's R type-6 quantiles. The three unavailable break durations comprise two structurally undefined no-break cases and one unresolved range. Grade proxies and ordinal ratings are summarized numerically for completeness, without assuming exact interval measurement. Administrative fields are not substantive variables.

Median session length is 120 minutes, with two breaks per session. Break duration has median 10 minutes, mean 16.89 and SD 18.81, indicating a long upper tail. Retention averages 6.15 (SD 1.40). Valid weekly study time averages 15.81 hours ($n = 43$). Similar means and medians alone do not establish normality.

Table 3a. Prior course grade band.

Category	n	% of 48
50-59%	1	2.1
60-69%	10	20.8
70-79%	25	52.1
80-89%	5	10.4
90-100%	3	6.2
Missing / unavailable	4	8.3

Table 3b. Break benefit agreement.

Category	n	% of 48
Strongly disagree	0	0.0
Disagree	2	4.2
Slightly disagree	0	0.0
Neutral	4	8.3
Slightly agree	8	16.7
Agree	23	47.9
Strongly agree	11	22.9
Missing / unavailable	0	0.0

The 70–79% prior-course grade band was the most common (52.1%). Forty-two respondents (87.5%) at least slightly agree that breaks help efficiency; this reflects perceived benefits rather than measured improvement. There are four respondents chose not prefer to say to indicate their prior course. Percentages use all 48 screened respondents, including missing cases; rounding may prevent exact totals of 100%.

3 Response distributions and visual exploration

Table 4a. Reported break activities.

Activity	n	% of 48
Eating	17	35.4
Exercise	9	18.8
Gaming	17	35.4
Napping	11	22.9
none	1	2.1
Instagram write-in	1	2.1
Scrolling	1	2.1
Scrolling Phone	1	2.1
Social	37	77.1
Space out	1	2.1
Youtube	1	2.1

Table 4b. Reported study locations.

Location	n	% of 48
Cafe	4	8.3
Home	41	85.4
Lecture hall	6	12.5
Library	35	72.9
Other campus building	2	4.2
Outdoors	1	2.1
Public transit	3	6.2
Missing / unusable	1	2.1

Multiple selections are counted once per respondent per label; percentages can exceed 100%. All selections are retained, including those exceeding the requested two. "Social" remains the source label without an assumed activity type. The Instagram write-in is shortened for display. Location splitting preserves commas inside parentheses, and campus-building aliases are combined. Activities have no missing responses; one location is unusable.

"Social" is selected by 77.1% of respondents; home (85.4%) and library (72.9%) are the most frequently reported locations. These categories overlap.

Table 4c. Distribution of perceived next-day retention.

Rating	1	2	3	4	5	6	7	8	9	10
Count	0	0	3	4	4	17	13	6	1	0
% of 48	0	0	6.2	8.3	8.3	35.4	27.1	12.5	2.1	0

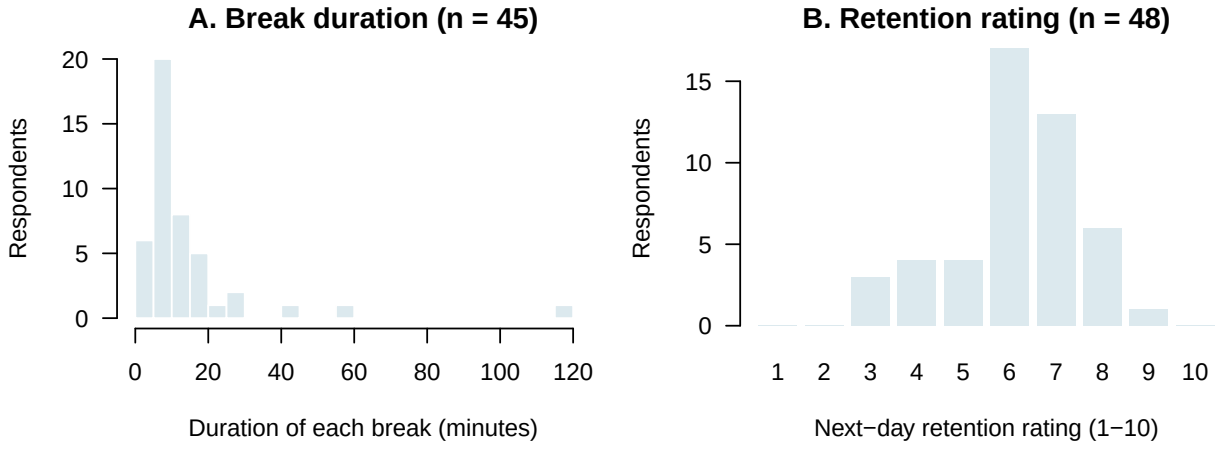


Figure 1. Distributions of the regression variables. A: observed break duration for the 45 model cases; the isolated 120-minute value creates a long right tail. B: retention for all 48 screened respondents, with 62.5% rating 6 or 7. Retention is bounded and discrete; residual diagnostics, rather than normality of the predictor, determine the suitability of normal-error inference.

4 Planned tests and RQ1: Chi-square independence test

The three proposal analyses are exploratory, each using $\alpha = 0.05$. Their nominal p-values are reported without a multiplicity correction; considering three questions increases the chance of at least one false positive. No additional inferential method is substituted when an assumption is violated.

Table 5. Tests specified in the latest proposal.

Question	Test	n	Statistic	Nominal p
RQ1	Pearson chi-square	48	19.93	0.0106
RQ2	Simple linear regression	45	-0.07	0.9445
RQ3	One-way ANOVA	46	0.98	0.4128

The statistics are respectively Pearson χ^2 (df 8), slope t (df 43), and ANOVA F (df 3, 42). The chi-square approximation is unreliable because of sparse expected counts. Regression inference also has assumption limitations.

Variables and operational choices. Following the proposal's Q2–Q9 comparison, break count is grouped as 0, 1, 2, 3 and 4+ per study session. The seven-point agreement rating is summarized as disagree (1–3), neutral (4) or agree (5–7). This pooling is an analysis choice, not a rule explicitly specified in the proposal. Break count is per session, not per hour; agreement measures perceived benefit, not directly observed efficiency.

Table 6. Observed counts for break count and benefit agreement ($n = 48$).

Breaks per session	Disagree (1–3)	Neutral (4)	Agree (5–7)
0	1	1	0
1	1	2	12
2	0	1	12
3	0	0	11
4+	0	0	7

Test H_0: break-count group and agreement group are independent, against association. Pearson's statistic is $\chi^2(8) = 19.925$, nominal $p = 0.0106$. However, **11 of 15 expected counts are below 5**, with minimum 0.083. This substantially violates the expected-count requirement for the asymptotic reference distribution. The numerical p-value is retained as the planned course analysis, but **cannot support a reliable significance claim**. No alternative test is introduced. Expected counts are exported to `tables/expected_counts.csv` for inspection.

Observed direction. Agreement occurs in 0/2 zero-break respondents, 12/15 one-break respondents, 12/13 two-break respondents, and all 18 respondents reporting three or more breaks. Higher counts therefore accompany more favorable beliefs descriptively. The zero-break group is extremely small, and this pattern does not establish an improvement in efficiency or a causal effect of breaks.

5 RQ2: Simple linear regression

Let X_i be break duration in minutes and Y_i the next-day retention rating. For the 45 complete break takers, fit

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \qquad H_0 : \beta_1 = 0 \quad \text{versus} \quad H_A : \beta_1 \neq 0.$$

The primary test is two-sided at $\alpha = 0.05$. It is one of the three exploratory tests planned in the proposal. The model treats the rating scale as approximately numeric.

Table 7. Ordinary least-squares coefficients and conventional 95% CIs.

Term	Estimate	SE	t	p	CI lower	CI upper
Intercept	6.21302	0.27670	22.45421	< 0.00001	5.65501	6.77103
Break duration (min)	-0.00077	0.01101	-0.07000	0.94452	-0.02298	0.02144

$n = 45$; residual df $= 43$; $R^2 = 0.000114$; adjusted $R^2 = -0.02314$; residual standard error $= 1.374$ rating points. Overall $F(1, 43) = 0.0049$, with the same p-value as the slope test. Conventional inference is approximate given the diagnostics in Section 7.

The fitted equation is

$$\widehat{\text{retention}} = 6.21302 + (-0.000771) \times \text{break minutes}.$$

The slope is -0.000771 rating points per minute (95% CI $[-0.02298, 0.02144]$), $t(43) = -0.070$, $p = 0.9445$. A 10-minute longer break corresponds to an estimated -0.0077 point difference (95% CI $[-0.2298, 0.2144]$). The fitted direction is slightly negative, but the slope is not statistically significant and the explained variation is less than 0.02%. This does not establish that the association is zero.

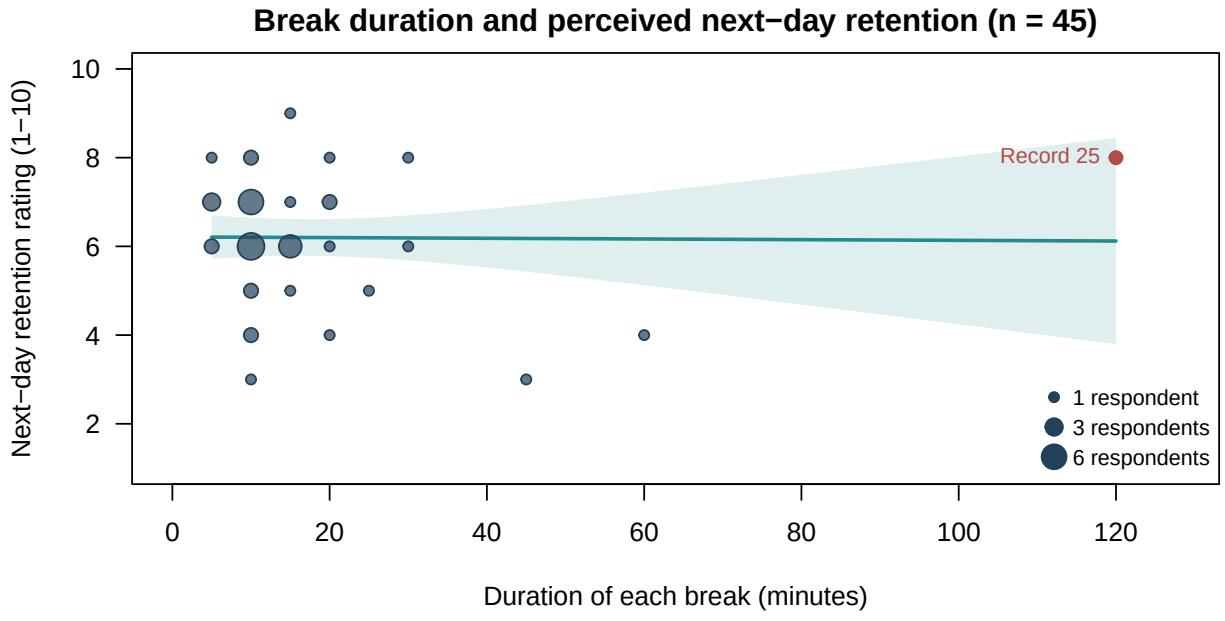


Figure 2. Retention versus duration and fitted regression. Point area represents the number of identical responses. The teal line is the fitted conditional mean; shading is its conventional pointwise 95% confidence band, not an interval for individual responses. The sparse upper-duration range and record 25 make the near-flat line sensitive to influence.

The intercept predicts a rating at zero minutes, outside the observed 5–120 minute range. It is not a comparison with students taking no breaks, who are excluded from this model because their break duration is undefined.

6 RQ3: One-way ANOVA of activity profiles

The proposal asks whether mean retention differs by activity type. The collected activity question allows multiple selections, so assigning one person to several independent groups would duplicate their outcome. We retain the previous work's four exclusive **exercise/gaming profiles**: exercise without gaming, gaming without exercise, neither, and both. Other activities may co-occur in every group. This is a narrower operational comparison than all possible break types and is an analysis choice, not a grouping prescribed by the proposal.

All 46 screened break takers enter this comparison, including one with missing break duration. Each contributes once. "Only" below refers to exercise versus gaming, not to the absence of all other activities. The ambiguous label "Social" is not recoded or used to assign these profiles.

Table 8. Perceived next-day retention by exclusive activity profile.

Profile	n	Mean	SD	Median
Exercise only	5	6.20	0.45	6.00
Gaming only	12	5.75	1.54	6.00
Neither	25	6.32	1.35	7.00
Both	4	7.00	1.41	7.50

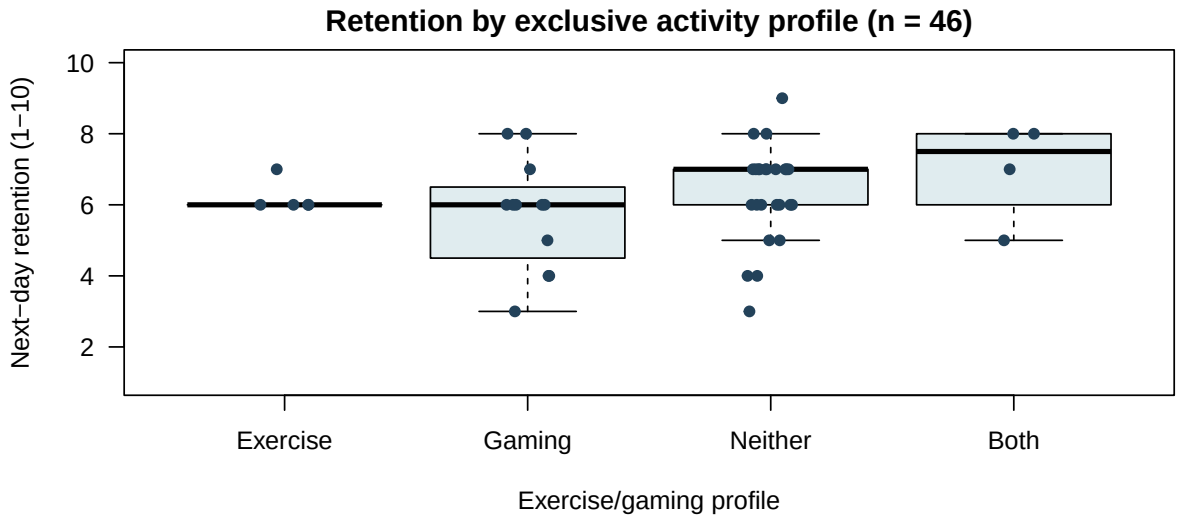


Figure 3. Retention across exclusive activity profiles. Boxes show medians and quartiles; points show individual observations with horizontal jitter only. Exercise and gaming labels indicate the corresponding exclusive profiles. The "both" group has the highest mean but only four respondents, and the distributions overlap considerably.

Test $H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$ against at least one unequal mean. The planned one-way ANOVA gives $F(3, 42) = 0.977$, $p = 0.4128$, with $\eta^2 = 0.065$. It does not reject equality at 0.05. The higher observed mean in the "both" profile is not a statistically established difference. No post-hoc comparisons or additional rank-based or unequal-variance tests are performed. This comparison does not isolate the effect of a single activity.

7 Model assumption and influence checks

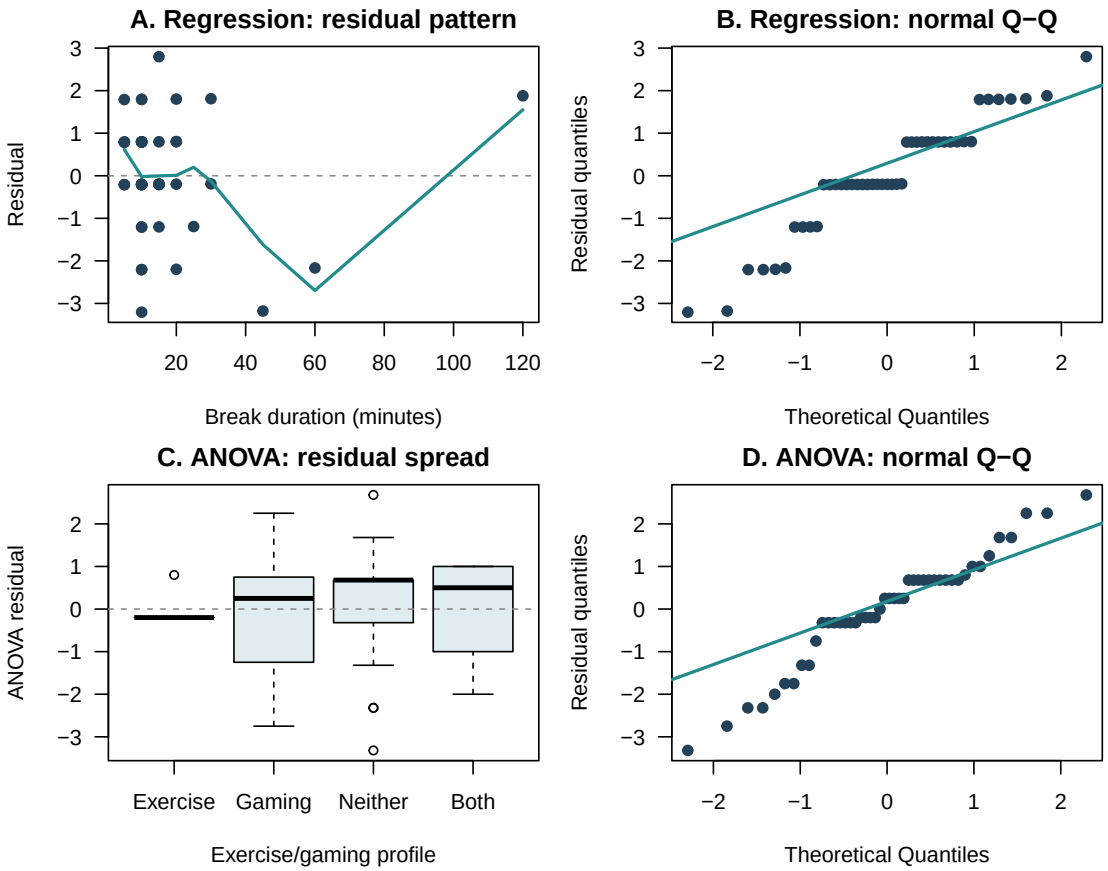


Figure 4. Diagnostics for the planned regression and ANOVA. A: regression residual pattern and spread against duration. B: its normal Q–Q plot. C: ANOVA residual spread across profiles. D: ANOVA normal Q–Q plot. Tied ratings and sparse long breaks limit the plots' resolution.

Table 9. Diagnostic tests supplementing the plots.

Model	Check	p
Regression	Shapiro-Wilk residual normality	0.00975
Regression	Variance score test	0.04998
ANOVA	Shapiro-Wilk residual normality	0.06311
ANOVA	Median-centered Levene test	0.43691

Regression. The residual trend is not uniformly flat, Q–Q tails depart from normality, and Shapiro–Wilk rejects ($p = 0.00975$). The variance score test is just below 0.05 ($p = 0.04998$), giving borderline evidence against equal variance. Conventional regression inference is consequently approximate. The retained 120-minute response (record 25) has leverage 0.705 and Cook's distance 7.60, indicating strong influence. It is not deleted to change significance, and no alternative regression or deletion test is added.

ANOVA. Neither normality nor median-centered Levene's test rejects at 0.05. This does not prove the assumptions, especially with groups of four and five. Independence is assumed from individual survey responses and exclusive profile membership; neither diagnostic plots nor ID screening can verify it. The course permits reporting violated assumptions while proceeding with the planned analysis; the results above are interpreted with those limitations.

8 Interpretation, alignment and limitations

RQ1. The observed pattern is toward more favorable break-benefit beliefs at higher break counts. Although the planned Pearson calculation gives a nominal $p = 0.0106$, sparse expected counts make this approximation unreliable. A firm significance conclusion is not warranted from that number.

RQ2. The fitted duration slope is near zero and not significant ($p = 0.9445$). Residual departures and one highly influential response limit inference. The result does not prove that duration has no relationship with retention and does not identify an optimal break length.

RQ3. The planned ANOVA does not find a significant mean difference across the four exclusive exercise/gaming profiles ($p = 0.4128$). The largest observed group mean is descriptive and is based on only four responses.

Alignment with the latest proposal. The only research hypothesis tests are chi-square independence, the simple-regression slope, and one-way ANOVA. Normality, variance and influence checks support the EDA requirements. The measurement update, category pooling and profile definition are disclosed rather than represented as original proposal specifications. A copy of the controlling proposal accompanies the source files. No older frequency–efficiency regression from the first proposal draft is used.

Limitations.

1. **Sampling and eligibility.** The planned SRS is not documented as implemented. Voluntary response and screening may introduce bias; the remaining course IDs are not roster-verified. Results concern the supplied respondents, not a validated random sample of the full course.
2. **Preparation.** The missing-ID and timing exclusions, structural missing durations for zero-break cases, and variable-level weekly-hour exclusions are transparent preparation choices. Flags alone do not prove invalidity. Unresolved ranges remain missing. These choices can affect the analysis sample.
3. **Measurement.** Retention is a bounded ordinal self-rating, not an objective test score; agreement is an attitude. Grades are band proxies, and the year prompt is problematic. Treating ratings numerically is an approximation.
4. **Attention-check failure.** The questionnaire has included an attention check question explicitly instructing respondents to select "First Year." However, there was only 5 respondents out of 48 selected the correct option.The high failure rate raises concerns about respondent attentiveness and the reliability of self-reported data. Respondents were not automatically excluded because failing the check does not necessarily invalidate their other answers. Moreover, this data cannot be interpreted as the actual study years of the respondents.
5. **Confounding and precision.** Session length, study load and personal differences are not adjusted for. Activity profiles contain co-occurring choices, small groups and an incomplete comparison of all break types. Sparse contingency cells and influential long breaks limit inferential precision. None of the three analyses supports a causal learning-improvement claim.

Reproducibility. Preparation, statistics and graphics use R and the easystats package `datawizard`. Inputs, screened data, exact regression cases, exported tables, figures and software versions accompany the report. Complete code appears in Appendix A. Numerical regression and test results in the text are populated by R-generated LaTeX macros.

References. Latest group project proposal (supplied as `proposal_reference.txt`); STA304 Exploratory Data Analysis Instructions (2026), course PDF/Quarto source. Patil et al. (2022), *datawizard: An R Package for Easy Data Preparation and Statistical Transformations*, JOSS 7(78), 4684, doi:10.21105/joss.04684. Package citation and software versions are included in `tables/datawizard_citation.txt` and `session_info.txt`.

A Complete reproducible R code

Run `Rscript analysis.R` from this folder after installing `datawizard`, `bayestestR`, `car` and `jsonlite`. The fixed input is `survey_source.csv`; source values are never overwritten.

```
# STA304 EDA. Run Rscript analysis.R from the submission folder.
# Input: unchanged copy of the new 57-record CSV and its codebook.
# All statistics, preparation and graphics are computed in R.
options(stringsAsFactors = FALSE, width = 78)
needed <- c("datawizard", "bayestestR", "car", "jsonlite")
if (!all(vapply(needed, requireNamespace, logical(1), quietly = TRUE))) {
  stop("Install the required packages: ", paste(needed, collapse = ", "))
}
dir.create("tables", showWarnings = FALSE)
dir.create("figures", showWarnings = FALSE)
raw <- read.csv("survey_source.csv", na.strings = "", check.names = FALSE,
  fileEncoding = "UTF-8-BOM", colClasses = c(studentID = "character"))
raw$record <- seq_len(nrow(raw))
raw$bad_id <- is.na(raw$studentID) | !nzchar(trimws(raw$studentID))
total <- with(raw, num_breaks * break_duration_min)
raw$bad_time <- !is.na(total) & total >= raw$study_session_min
raw$bad_week <- !is.na(raw$weekly_study_hours) &
  (raw$weekly_study_hours > 168 | raw$weekly_study_hours < 0)
raw$excluded <- raw$bad_id | raw$bad_time
d <- raw[!raw$excluded, ]
d$weekly_study_hours[d$bad_week] <- NA_real_
# No-break respondents have no defined duration of an actual break.
# Do not infer a duration of zero from the contradictory positive answer.
d$break_duration_min[d$num_breaks == 0] <- NA_real_
split_multi <- function(x) {
  lapply(x, function(z) if (is.na(z)) character() else
    strsplit(z, ",\\s*(?![^()]*\\))", perl = TRUE)[[1]])
}
activities <- split_multi(d$break_activities)
has <- function(label) vapply(activities, function(z) label %in% z, logical(1))
labels <- c(study_session_min = "Study session duration (min)",
  num_breaks = "Breaks per study session",
  break_duration_min = "Duration of each break (min)",
  memory_retention = "Next-day retention rating (1-10)",
  weekly_study_hours = "Weekly study time (hours)",
  prior_grade_pct = "Prior grade: band proxy (%)",
  break_helps_efficiency = "Break benefit agreement (1-7)")
ds <- as.data.frame(datawizard::describe_distribution(d[names(labels)],
  centrality = c("mean", "median")))
write.csv(ds, "tables/easystats_summary.csv", row.names = FALSE)
num <- data.frame(Variable = unname(labels[ds$Variable]), n = ds$n,
  Missing = ds$n_Missing, Mean = ds$Mean, Median = ds$Median,
  SD = ds$SD, IQR = ds$IQR,
  Min = vapply(d[ds$Variable], min, numeric(1), na.rm = TRUE),
  Max = vapply(d[ds$Variable], max, numeric(1), na.rm = TRUE))
freq <- function(x, levels, labels = levels) {
  z <- table(factor(x, levels = levels), useNA = "always")
  data.frame(Category = c(labels, "Missing / unavailable"),
    n = as.integer(z), Percent = 100 * as.integer(z) / length(x))
}
year <- freq(d$year_of_study, 1:5, paste("Year", 1:5))
grade <- freq(d$prior_grade_pct, c(55, 65, 75, 85, 95),
  c("50-59%", "60-69%", "70-79%", "80-89%", "90-100%"))
agree <- freq(d$break_helps_efficiency, 1:7,
  c("Strongly disagree", "Disagree", "Slightly disagree", "Neutral",
    "Slightly agree", "Agree", "Strongly agree"))
```

```
memory <- freq(d$memory_retention, 1:10, as.character(1:10))
al <- sort(unique(unlist(activities)))
act <- data.frame(Category = al,
  n = vapply(al, function(z) sum(has(z)), integer(1)))
act$Category[act$Category == "Rotting on instagram specifically"] <-
  "Instagram write-in"
act$Percent <- 100 * act$n / nrow(d)
loc <- split_multi(d$study_location)
loc <- lapply(loc, function(z) {
  z[z %in% c("UTM buildings like IB or MN",
    "University Buildings (e.g. MN)")] <- "Other campus building"
  z[z == "Lecture hall(but self-study in lecture hall)"] <- "Lecture hall"
  z[z == "Transit(in bus, subway, train, etc)"] <- "Public transit"
  z[z == "Outdoors(park, beach, etc)"] <- "Outdoors"
  unique(z)
})
ll <- sort(unique(unlist(loc)))
location <- data.frame(Category = c(ll, "Missing / unusable"),
  n = c(vapply(ll, function(z) sum(vapply(loc,
    function(v) z %in% v, logical(1))), integer(1)),
    sum(lengths(loc) == 0)))
location$Percent <- 100 * location$n / nrow(d)
ex <- has("Exercise"); ga <- has("Gaming")
d$activity_profile <- factor(ifelse(ex & ga, "Both",
  ifelse(ex, "Exercise only", ifelse(ga, "Gaming only", "Neither"))),
  levels = c("Exercise only", "Gaming only", "Neither", "Both"))
d$break_count_group <- cut(d$num_breaks, c(-1, 0, 1, 2, 3, Inf),
  labels = c("0", "1", "2", "3", "4+"))
d$agreement_group <- cut(d$break_helps_efficiency, c(0, 3, 4, 7),
  labels = c("Disagree (1-3)", "Neutral (4)", "Agree (5-7)"))
breakers <- droplevels(d[d$num_breaks > 0, ])
bt <- breakers[complete.cases(breakers[c("break_duration_min",
  "memory_retention")]), ]
ad <- droplevels(breakers[!is.na(breakers$memory_retention), ])
# Only the three analyses listed in proposal_reference.txt are fitted below.
cross <- with(d, table(break_count_group, agreement_group))
# Preserve the planned Pearson test. Sparse expected counts are explicitly
# reported as a limitation; no replacement test is substituted.
chi <- suppressWarnings(chisq.test(cross, correct = FALSE))
ols <- lm(memory_retention ~ break_duration_min, data = bt)
av <- aov(memory_retention ~ activity_profile, data = ad)
sm <- summary(ols); avtab <- summary(av)[[1]]; ci <- confint(ols)
sh_ols <- shapiro.test(residuals(ols)); ncv <- car::ncvTest(ols)
sh_av <- shapiro.test(residuals(av))
lev <- car::leveneTest(memory_retention ~ activity_profile,
  data = ad, center = median)
coeff <- data.frame(Term = c("Intercept", "Break duration (min)"),
  Estimate = unname(coef(ols)), SE = sm$coefficients[, 2],
  t = sm$coefficients[, 3], p = sm$coefficients[, 4],
  Lower = ci[, 1], Upper = ci[, 2], row.names = NULL)
fit <- data.frame(n = nrow(bt), Residual_df = df.residual(ols),
  R_squared = sm$r.squared, Adjusted_R_squared = sm$adj.r.squared,
  Residual_SE = sm$sigma, F = unname(sm$fstatistic[1]))
tests <- data.frame(Question = c("RQ1", "RQ2", "RQ3"),
  Test = c("Pearson chi-square", "Simple linear regression", "One-way ANOVA"),
  n = c(nrow(d), nrow(bt), nrow(ad)),
  Statistic = c(unname(chi$statistic), sm$coefficients[2, 3],
    avtab[1, "F value"]),
  p = c(chi$p.value, sm$coefficients[2, 4], avtab[1, "Pr(>F)"]))
groups <- do.call(rbind, lapply(split(ad$memory_retention,
  ad$activity_profile), function(z) c(n = length(z), Mean = mean(z),
```

```
    SD = sd(z), Median = median(z))))
groups <- data.frame(Profile = rownames(groups), groups, row.names = NULL)
checks <- data.frame(Model = c("Regression", "Regression", "ANOVA", "ANOVA"),
  Check = c("Shapiro-Wilk residual normality", "Variance score test",
    "Shapiro-Wilk residual normality", "Median-centered Levene test"),
  p = c(sh_ols$p.value, ncv$p, sh_av$p.value, lev[1, "Pr(>F)"]))
cooks <- cooks.distance(ols)
influence <- data.frame(record = bt$record, fitted = fitted(ols),
  residual = residuals(ols), standardized_residual = rstandard(ols),
  leverage = hatvalues(ols), cooks_distance = cooks)
safe <- d[setdiff(names(d), c("studentID", "timestamp", "data_notes"))]
write.csv(safe, "analysis_ready.csv", row.names = FALSE, na = "")
write.csv(bt[c("record", "break_duration_min", "memory_retention")],
  "regression_data.csv", row.names = FALSE)
write.csv(raw[c("record", "bad_id", "bad_time", "bad_week", "excluded")],
  "tables/record_audit.csv", row.names = FALSE)
write.csv(influence, "tables/regression_influence.csv", row.names = FALSE)
write.csv(fit, "tables/model_fit.csv", row.names = FALSE)
write.csv(as.data.frame.matrix(chi$expected),
  "tables/expected_counts.csv", row.names = TRUE)
all_tables <- list(numeric = num, year = year, grade = grade,
  agreement = agree, retention = memory, activities = act,
  locations = location, model_coefficients = coeff, primary_tests = tests,
  group_summary = groups, assumption_checks = checks)
tex_escape <- function(x) {
  x <- gsub("%", "\\%", x, fixed = TRUE)
  x <- gsub("&", "\\&", x, fixed = TRUE)
  gsub("_", "\\_", x, fixed = TRUE)
}
for (nm in names(all_tables)) {
  tb <- all_tables[[nm]]
  write.csv(tb, paste0("tables/", nm, ".csv"), row.names = FALSE)
  display <- tb
  for (cn in names(tb)) {
    if (is.numeric(tb[[cn]])) {
      dp <- if (cn %in% c("n", "Missing")) 0 else
        if (cn == "Percent") 1 else
        if (nm %in% c("model_coefficients", "assumption_checks")) 5 else
        if (cn == "p") 4 else 2
      display[[cn]] <- formatC(tb[[cn]], format = "f", digits = dp)
      if (cn == "p") display[[cn]][tb[[cn]] < .00001] <- "$<0.00001$"
    }
    display[[cn]] <- tex_escape(display[[cn]])
  }
  lines <- apply(display, 1, function(z)
    paste0(paste(z, collapse = " & "), " \\\\"))
  writeLines(c(lines, "\\bottomrule"), paste0("tables/", nm, ".tex"))
}
ct <- data.frame(Breaks = rownames(cross), unclass(cross), check.names = FALSE)
write.csv(ct, "tables/contingency.csv", row.names = FALSE)
ct_lines <- apply(ct, 1, function(z)
  paste0(paste(z, collapse = " & "), " \\\\"))
writeLines(c(ct_lines, "\\bottomrule"), "tables/contingency.tex")
values <- c(Intercept = sprintf("%.5f", coef(ols)[1]),
  Slope = sprintf("%.6f", coef(ols)[2]),
  SlopeP = sprintf("%.4f", coeff$p[2]),
  SlopeT = sprintf("%.3f", coeff$t[2]),
  SlopeLo = sprintf("%.5f", ci[2, 1]),
  SlopeHi = sprintf("%.5f", ci[2, 2]),
  RSquared = sprintf("%.6f", sm$r.squared),
  AdjRSquared = sprintf("%.5f", sm$adj.r.squared),
```

```
  ResidualSE = sprintf("%.3f", sm$sigma),
  FStat = sprintf("%.4f", unname(sm$fstatistic[1])),
  TenEffect = sprintf("%.4f", 10 * coef(ols)[2]),
  TenLo = sprintf("%.4f", 10 * ci[2, 1]),
  TenHi = sprintf("%.4f", 10 * ci[2, 2]),
  ShapiroP = sprintf("%.5f", sh_ols$p.value),
  NCVP = sprintf("%.5f", ncv$p),
  MaxCook = sprintf("%.2f", max(cooks)),
  MaxHat = sprintf("%.3f", max(hatvalues(ols))),
  ChiStat = sprintf("%.3f", chi$statistic),
  ChiP = sprintf("%.4f", chi$p.value),
  ExpectedMin = sprintf("%.3f", min(chi$expected)),
  AnovaF = sprintf("%.3f", avtab[1, "F value"]),
  AnovaP = sprintf("%.4f", avtab[1, "Pr(>F)"]),
  EtaSquared = sprintf("%.3f", avtab[1, "Sum Sq"] / sum(avtab[, "Sum Sq"])))
names(values) <- sub("\\..*$", "", names(values))
writeLines(paste0("\\newcommand{\\", names(values), "}{", values, "}"),
  "tables/report_values.tex")
navy <- "#23425A"; teal <- "#238A8D"; red <- "#B44B45"
pdf("figures/descriptive_distributions.pdf", width = 7.2, height = 2.65)
par(mfrow = c(1, 2), mar = c(4.1, 3.9, 2.2, .7), cex = .82, las = 1)
hist(bt$break_duration_min, breaks = seq(0, 125, 5), col = "#DCE9EE",
  border = "white", main = "A. Break duration (n = 45)",
  xlab = "Duration of each break (minutes)", ylab = "Respondents")
barplot(as.integer(table(factor(d$memory_retention, levels = 1:10))),
  names.arg = 1:10, col = "#DCE9EE", border = NA,
  main = "B. Retention rating (n = 48)",
  xlab = "Next-day retention rating (1-10)", ylab = "Respondents")
dev.off()
pdf("figures/research_results.pdf", width = 7.2, height = 3.65)
par(mar = c(4.2, 4.2, 2.1, .8), cex = .92, las = 1)
grid <- data.frame(break_duration_min = seq(min(bt$break_duration_min),
  max(bt$break_duration_min), length.out = 250))
prediction <- predict(ols, grid, interval = "confidence")
plot(NA, xlim = c(0, 128), ylim = c(1, 10),
  xlab = "Duration of each break (minutes)",
  ylab = "Next-day retention rating (1-10)",
  main = "Break duration and perceived next-day retention (n = 45)")
polygon(c(grid$break_duration_min, rev(grid$break_duration_min)),
  c(prediction[, "lwr"], rev(prediction[, "upr"])),
  col = adjustcolor(teal, alpha.f = .15), border = NA)
lines(grid$break_duration_min, prediction[, "fit"], col = teal, lwd = 2)
points_n <- aggregate(rep(1, nrow(bt)),
  list(Minutes = bt$break_duration_min, Rating = bt$memory_retention), sum)
points(points_n$Minutes, points_n$Rating, pch = 21,
  bg = adjustcolor(navy, alpha.f = .7), col = navy,
  cex = .85 * sqrt(points_n$x))
im <- which.max(cooks)
points(bt$break_duration_min[im], bt$memory_retention[im],
  pch = 21, col = red, bg = red, cex = 1.1)
text(bt$break_duration_min[im], bt$memory_retention[im],
  labels = "Record 25", pos = 2, col = red, cex = .83)
legend("bottomright", c("1 respondent", "3 respondents", "6 respondents"),
  pch = 21, pt.bg = navy, col = navy, pt.cex = .85 * sqrt(c(1, 3, 6)),
  bty = "n", cex = .78, y.intersp = 1.2)
dev.off()
short <- c("Exercise", "Gaming", "Neither", "Both")
pdf("figures/activity_comparison.pdf", width = 7.2, height = 3.2)
par(mar = c(4.2, 4, 2, .8), cex = .9, las = 1)
boxplot(memory_retention ~ activity_profile, data = ad, names = short,
  col = "#E0ECEF", xlab = "Exercise/gaming profile",
```

```
  ylab = "Next-day retention (1-10)", ylim = c(1, 10),
  main = "Retention by exclusive activity profile (n = 46)", outline = FALSE)
set.seed(304)
stripchart(memory_retention ~ activity_profile, data = ad,
  method = "jitter", vertical = TRUE, add = TRUE, pch = 19, col = navy)
dev.off()
pdf("figures/model_diagnostics.pdf", width = 7.2, height = 5.6)
par(mfrow = c(2, 2), mar = c(4, 4, 2.1, 1), cex = .85, las = 1)
plot(bt$break_duration_min, residuals(ols), pch = 19, col = navy,
  xlab = "Break duration (minutes)", ylab = "Residual",
  main = "A. Regression: residual pattern")
abline(h = 0, col = "grey55", lty = 2)
lines(lowess(bt$break_duration_min, residuals(ols)), col = teal, lwd = 2)
qqnorm(residuals(ols), pch = 19, col = navy,
  main = "B. Regression: normal Q-Q", ylab = "Residual quantiles")
qqline(residuals(ols), col = teal, lwd = 2)
boxplot(residuals(av) ~ ad$activity_profile, names = short, col = "#E0ECEF",
  xlab = "Exercise/gaming profile", ylab = "ANOVA residual",
  main = "C. ANOVA: residual spread")
abline(h = 0, col = "grey55", lty = 2)
qqnorm(residuals(av), pch = 19, col = navy,
  main = "D. ANOVA: normal Q-Q", ylab = "Residual quantiles")
qqline(residuals(av), col = teal, lwd = 2)
dev.off()
pdf("figures/data_quality.pdf", width = 7.2, height = 3.35)
par(mfrow = c(1, 2), mar = c(4, 4, 2.2, 1), cex = .84, las = 1)
ok <- !is.na(total) & total > 0
plot(raw$study_session_min[ok], total[ok], log = "xy", pch = 19,
  col = ifelse(raw$bad_time[ok], red, navy),
  xlab = "Session minutes (log scale)",
  ylab = "Total break minutes (log scale)", main = "A. Session-time consistency")
abline(0, 1, col = "grey55", lty = 2)
extreme <- which(raw$bad_time & total > 1000)
text(raw$study_session_min[extreme], total[extreme],
  labels = paste("Record", extreme), pos = 2, cex = .8)
wk <- !is.na(raw$weekly_study_hours) & raw$weekly_study_hours > 0
plot(raw$record[wk], raw$weekly_study_hours[wk], log = "y", pch = 19,
  col = ifelse(raw$bad_week[wk], red, navy), xlab = "Source record number",
  ylab = "Weekly study time (hours; log scale)", main = "B. Weekly-time outliers")
abline(h = 168, lty = 2, col = red)
text(18, 168, "168 hours/week", pos = 3, cex = .8, col = red)
dev.off()
result <- list(method_reference = "proposal_reference.txt",
  raw_n = nrow(raw), screened_n = nrow(d),
  break_takers_n = nrow(breakers), regression_n = nrow(bt), anova_n = nrow(ad),
  bad_id_records = raw$record[raw$bad_id],
  timing_records = raw$record[raw$bad_time],
  weekly_outlier_records = raw$record[raw$bad_week],
  numeric_summary = num, coefficients = coeff, model_fit = fit,
  primary_tests = tests, activity_groups = groups, assumptions = checks,
  chi_square = list(df = unname(chi$parameter),
    expected_min = min(chi$expected), expected_under5 = sum(chi$expected < 5),
    expected_cells = length(chi$expected), approximation_unreliable = TRUE),
  anova = list(df1 = avtab[1, "Df"], df2 = avtab[2, "Df"],
    eta_squared = avtab[1, "Sum Sq"] / sum(avtab[, "Sum Sq"])),
  influence = list(max_cook_record = bt$record[im],
    max_cook = max(cooks), max_leverage = max(hatvalues(ols))))
jsonlite::write_json(result, "results.json", auto_unbox = TRUE,
  pretty = TRUE, digits = 15)
capture.output(sessionInfo(), file = "session_info.txt")
capture.output(citation("datawizard"), file = "tables/datawizard_citation.txt")
```

B Source-data consistency and outlier plots

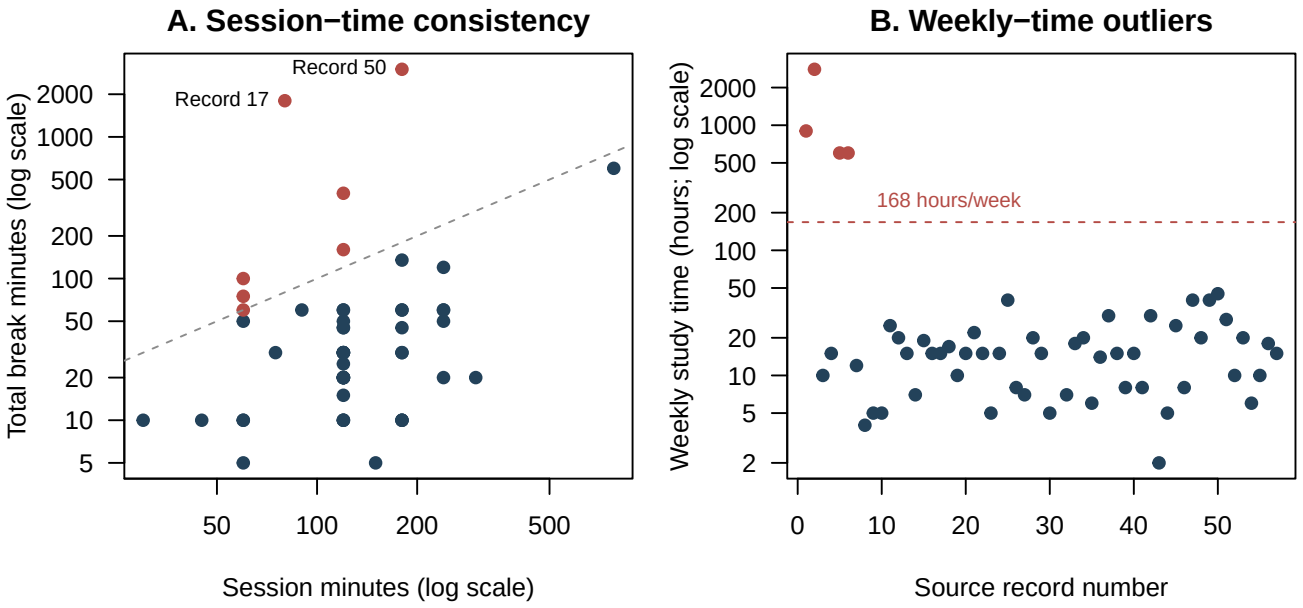


Figure B1. Source-data consistency checks. A: the 54 positive, observed total break times against session duration on log scales. The seven red points meet or exceed the equality line and are excluded by the timing screen. Two no-break zero totals and one missing duration are not shown on the log plot. Only the two largest inconsistent totals are labelled. B: the 56 observed weekly-hour entries, including four above the 168-hour limit. Those four fields are set to missing, without assuming a unit conversion. Source-record numbers start from 1 after the CSV header.